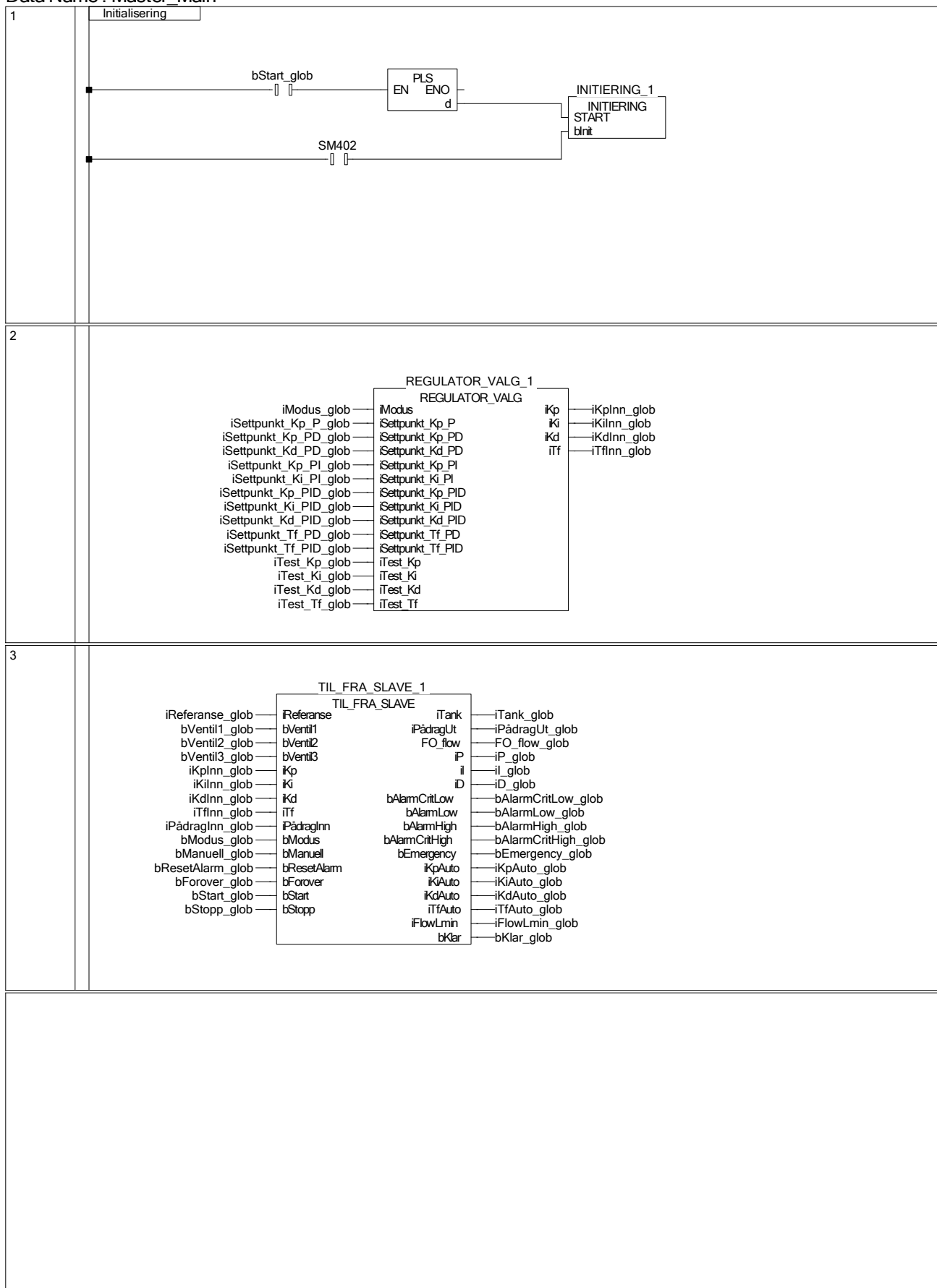
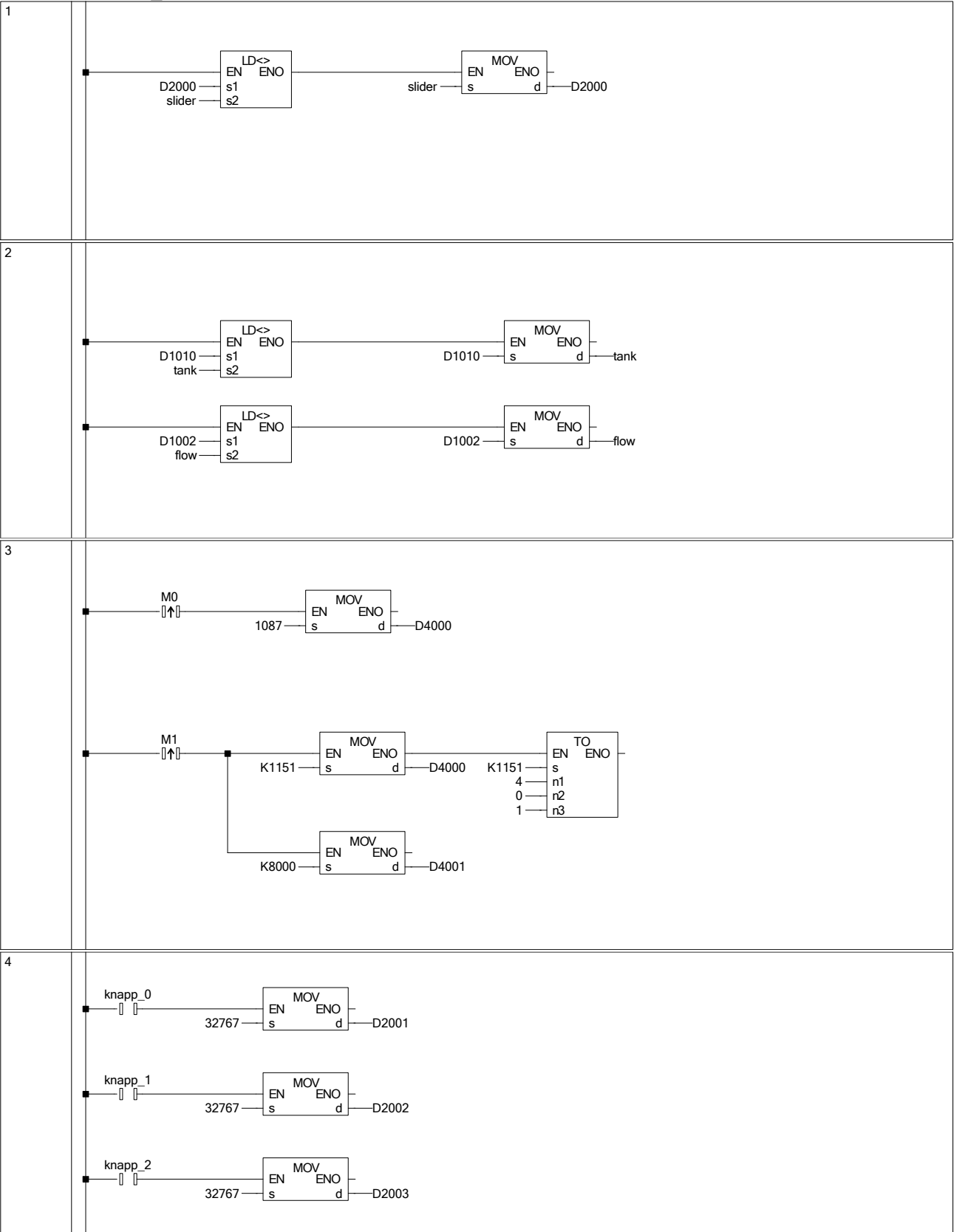
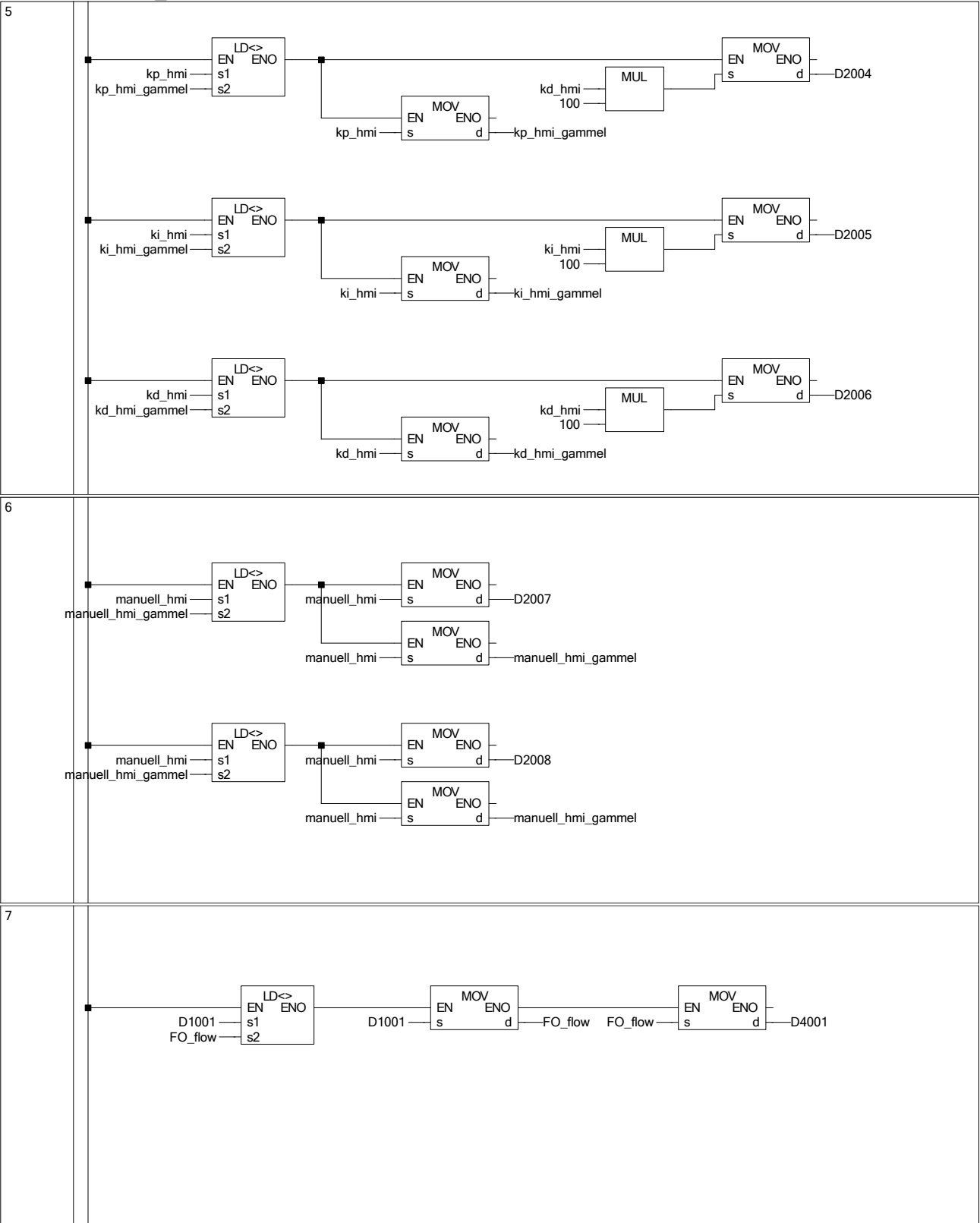








FB/FUN Program
Data Name : INITIERING
Function Block

5/3/2026

```
SET(binit, Y00);
OUT_T(Y00, TC0, 50);
MOV(TS0, 1087, D4000);

IF START THEN
  iKplnn_glob := 2470;
  iKlInn_glob := 78;
  iKdInn_glob := 5574;
  iTfInn_glob := 451;

  iRateSek := 5;

  iModus_glob := 6;
  iModus_saved_glob := 6;

  iSettpunkt_Kp_P_glob := 2000;

  iSettpunkt_Kp_PD_glob := 2200;
  iSettpunkt_Kd_PD_glob := 4500;
  iSettpunkt_Tf_PD_glob := 400;

  iSettpunkt_Kp_PI_glob := 1800;
  iSettpunkt_Ki_PI_glob := 60;

  iSettpunkt_Kp_PID_glob := 2470;
  iSettpunkt_Ki_PID_glob := 78;
  iSettpunkt_Kd_PID_glob := 5574;
  iSettpunkt_Tf_PID_glob := 451;

  iSettpunkt_Tf_PID_glob := 100;

  iReferanse_glob := 50;

  bModus_glob := FALSE;

  bForover_glob := FALSE;

  iNomPådrag := 0;

  STARTPREV := TRUE;
END_IF;

(* Bytte av pådragsorgan *)
IF bModus_glob AND NOT bModus_prev THEN
  iRateSek := 5;
  iSettpunkt_Kp_P_glob := 2000;
  iSettpunkt_Kp_PD_glob := 2200;
  iSettpunkt_Kd_PD_glob := 4500;
  iSettpunkt_Tf_PD_glob := 400;
  iSettpunkt_Kp_PI_glob := 1800;
  iSettpunkt_Ki_PI_glob := 60;
  iSettpunkt_Kp_PID_glob := 2470;
  iSettpunkt_Ki_PID_glob := 78;
  iSettpunkt_Kd_PID_glob := 5574;
  iSettpunkt_Tf_PID_glob := 451;
  iSettpunkt_NomPådrag_glob := 1300;
  IF ABS(Tank_glob - iReferanse_glob) < 50 AND il_glob > 2000 THEN
    iSettpunkt_NomPådrag_glob := il_glob;
  ELSE
    iSettpunkt_NomPådrag_glob := 1300;
  END_IF;
END_IF;

IF NOT bModus_glob AND bModus_prev THEN
  iRateSek := 5;
  iSettpunkt_Kp_P_glob := 2000;
  iSettpunkt_Kp_PD_glob := 2200;
  iSettpunkt_Kd_PD_glob := 4500;
  iSettpunkt_Tf_PD_glob := 400;
  iSettpunkt_Kp_PI_glob := 1800;
  iSettpunkt_Ki_PI_glob := 60;
  iSettpunkt_Kp_PID_glob := 2470;
  iSettpunkt_Ki_PID_glob := 78;
  iSettpunkt_Kd_PID_glob := 5574;
  iSettpunkt_Tf_PID_glob := 451;
  iSettpunkt_NomPådrag_glob := 3900;
  IF ABS(Tank_glob - iReferanse_glob) < 50 AND il_glob > 2000 THEN
    iSettpunkt_NomPådrag_glob := il_glob;
  ELSE
    iSettpunkt_NomPådrag_glob := 3800;
  END_IF;
END_IF;

bModus_prev := bModus_glob;
```

FB/FUN Program
Data Name : TIL_FRA_SLAVE
Function Block

5/3/2026

(* ===== FRA SLAVE ===== *)

(* Tank-nivå *)
iTank:= D1010;

(* Pådrag *)
MOV(TRUE, D1002, iPdragUt);

(* Frekvensomformer pådrag *)
IF bModus AND iPdragUt = 0 THEN
 MOV(TRUE, 2000, D4001);
ELSIF D1003 <> FO_flow THEN
 MOV(TRUE, D1003, FO_flow);
 MOV(TRUE, FO_flow, D4001);
END_IF;

(* Pådrag Kp *)
IF D1004 <> iP THEN
 MOV(TRUE, D1004, iP);
END_IF;

(* Pådrag Ki *)
IF D1005 <> iI THEN
 MOV(TRUE, D1005, iI);
END_IF;

(* Pådrag Kd *)
IF D1006 <> iD THEN
 MOV(TRUE, D1006, iD);
END_IF;

(* Alarm *)
bAlarmCritLow := D1007.0;
bAlarmLow := D1007.1;
bAlarmCritHigh := D1007.2;
bAlarmHigh := D1007.3;
bEmergency := D1007.4;
bManuelStatus_glob := D1007.5;
bAlarmFerdigStatus_glob := D1007.6;
bRunningStatus_glob := D1007.7;

bVentil1Status_glob := D1007.8;
bVentil2Status_glob := D1007.9;
bVentil3Status_glob := D1007.A;

IF bEmergency OR bRunningStatus_glob OR bStopp_glob THEN
 bVentil1_glob := bVentil1Status_glob;
 bVentil2_glob := bVentil2Status_glob;
 bVentil3_glob := bVentil3Status_glob;
END_IF;

IF NOT bEmergency_glob AND bEmergency_prev THEN
 bVentil1_glob := TRUE;
 bVentil2_glob := TRUE;
 bVentil3_glob := FALSE;
END_IF;
bEmergency_prev := bEmergency_glob;

IF bStart_glob AND NOT bStart_prev THEN
 bVentil1_glob := TRUE;
 bVentil2_glob := TRUE;
 bVentil3_glob := FALSE;
END_IF;
bStart_prev := bStart_glob;

(* Fra autotuning *)
IF D1011 <> iKpAuto THEN
 MOV(TRUE, D1011, iKpAuto);
 bAutoTuneFerdig_glob := 1;
END_IF;
OUT_T(bAutoTuneFerdig_glob, TC2, 50);
IF TS2 THEN
 bAutoTuneFerdig_glob := 0;
END_IF;

IF D1012 <> iKiAuto THEN
 MOV(TRUE, D1012, iKiAuto);
END_IF;

IF D1013 <> iKdAuto THEN
 MOV(TRUE, D1013, iKdAuto);
END_IF;

IF D1014 <> iTfAuto THEN
 MOV(TRUE, D1014, iTfAuto);
END_IF;

(* Oppstart *)
bKlar := D1015.1;

FB/FUN Program
Data Name : TIL_FRA_SLAVE
Function Block

5/3/2026

```

iReferanseAkt_glob := D1000;

(* Måling av utstrømning *)
IF D1001 <> iFlowLmin THEN
  MOV(TRUE, D1001, iFlowLmin);
END_IF;

(* Flanke: auto -> manuell *)
IF bManuelStatus_glob AND NOT bManuelStatus_prev THEN
  IF iPådragUt_glob >= 0 AND iPådragUt_glob <= 100 THEN
    iPådragInn_glob := iPådragUt_glob * 100;
  END_IF;
END_IF;
bManuelStatus_prev := bManuelStatus_glob;

(* ===== TIL SLAVE===== *)

(* Referanse *)
MOV(TRUE, iReferanse, D2000);

(* Ventil 1 *)
D2001.0 := bVentil1;
D2001.1 := bVentil2;
D2001.2 := bVentil3;

(* Kp *)
IF iKp <> iKp_old OR bModusEndret_glob THEN
  MOV(TRUE, iKp, D2004);
  MOV(TRUE, iKp, iKp_old);
END_IF;

(* Ki *)
IF iKi <> iKi_old OR bModusEndret_glob THEN
  MOV(TRUE, iKi, D2005);
  MOV(TRUE, iKi, iKi_old);
END_IF;

(* Kd *)
IF iKd <> iKd_old OR bModusEndret_glob THEN
  MOV(TRUE, iKd, D2006);
  MOV(TRUE, iKd, iKd_old);
END_IF;

(* Tf *)
IF iTf <> iTf_old OR bModusEndret_glob THEN
  MOV(TRUE, iTf, D2008);
  MOV(TRUE, iTf, iTf_old);
END_IF;

(* Manuell input *)
IF iPådragInn <> iPådrag_old THEN
  MOV(TRUE, iPådragInn, D2007);
  MOV(TRUE, iPådragInn, iPådrag_old);
END_IF;

(* Frekvensomformer eller ventil *)
D2001.3 := bModus;

(* Manuell eller auto modus *)
D2001.4 := bManuell;

(* Kvittering av alarm *)
IF bResetAlarm AND NOT bResetAlarm_old THEN
  bResetPuls := TRUE;
END_IF;
bResetAlarm_old := bResetAlarm;

OUT_T(bResetPuls, TC3, 20);
IF TS3 THEN
  bResetPuls := FALSE;
  bResetAlarm_glob := FALSE;
END_IF;
D2001.5 := bResetPuls;

(* Bruk foroverkobling *)
D2001.6 := bForover;

(* START/STOPP *)
D2001.7 := bStart;
D2001.8 := bStopp;

(* Autotuning *)
MOV(TRUE, D26, D2012);
D2012.5 := bResetAutotuning_glob;

(* Norm Pådrag *)

```

MOV(TRUE, D27, D2013); (* Foroverkoblingskoeffisient *) MOV(TRUE, D28, D2014); (* Ratebegrensning *) MOV(TRUE, D29, D2015);

```
oneshot:= Input AND NOT OldInput;  
  
IF oneshot THEN  
    FlipFlop := NOT FlipFlop;  
END_IF;  
  
OldInput:=Input;
```


FB/FUN Program
Data Name : REGULATOR_VALG
Function Block

5/3/2026

```
(* Lagre modus ved alarm *)
IF bEmergency_glob AND NOT bEmergency_prev THEN
  iModus_saved_glob := iModus_glob;
  iReferanse_glob := 50;
END_IF;

(* Gjenopprett modus etter reset *)
IF NOT bEmergency_glob AND bEmergency_prev THEN
  iModus_glob := iModus_saved_glob;
END_IF;
bEmergency_prev := bEmergency_glob;

(* Auto-bytte til PID når autotuning er ferdig *)
IF bAutoTuneFerdig_glob AND NOT bAutoTuneFerdig_prev THEN
  iModus_glob := 6;
  iModus_saved_glob := 6;
END_IF;
bAutoTuneFerdig_prev := bAutoTuneFerdig_glob;

bModusEndret := (iModus <> iModus_prev);
bModusEndret_glob := bModusEndret;
iModus_prev := iModus;

bOrganEndret := (bModus_glob <> bModus_prev_rv);
bModus_prev_rv := bModus_glob;

CASE iModus OF
  0:
    bManuell_glob := TRUE;
    iKp := iSettpunkt_Kp_PID;
    iKi := iSettpunkt_Ki_PID;
    iKd := iSettpunkt_Kd_PID;
    iTf := iSettpunkt_Tf_PID;

  2: (* Justerbar *)
    bManuell_glob := FALSE;
    iKp := iTTest_Kp;
    iKi := iTTest_Ki;
    iKd := iTTest_Kd;
    iTf := iTTest_Tf;

  3: (* P *)
    bManuell_glob := FALSE;
    iKp := iSettpunkt_Kp_P;
    iKi := 0;
    iKd := 0;
    iTf := 0;

  4: (* PD *)
    bManuell_glob := FALSE;
    iKp := iSettpunkt_Kp_PD;
    iKi := 0;
    iKd := iSettpunkt_Kd_PD;
    iTf := iSettpunkt_Tf_PID;

  5: (* PI *)
    bManuell_glob := FALSE;
    iKp := iSettpunkt_Kp_Pi;
    iKi := iSettpunkt_Ki_Pi;
    iKd := 0;
    iTf := 0;

  6: (* PID *)
    bManuell_glob := FALSE;
    iKp := iSettpunkt_Kp_PID;
    iKi := iSettpunkt_Ki_PID;
    iKd := iSettpunkt_Kd_PID;
    iTf := iSettpunkt_Tf_PID;
END_CASE;
```

Label
Data Name : Global1
Global Label Setting

5/3/2026

	Class	Label Name	Data Type	Constant	Device	Address	Comment	Remark	Relation with System Label	System Label Name	Attribute
1	VAR_GLOBAL	REGULATOR_VALG_1	REGULATOR_VALG								
2	VAR_GLOBAL	INTERRING_1	INTERRING								
3	VAR_GLOBAL	TL_FRA_SLAVE_1	TL_FRA_SLAVE								
4	VAR_GLOBAL	ITank_glob	Word(Signed)		D0	%MW0.0					
5	VAR_GLOBAL	IPdragU_glob	Word(Signed)		D10	%MW0.10	generert pdrag				
6	VAR_GLOBAL	PO_flow_glob	Word(Signed)		D11	%MW0.11	pdrag fra frekvensomformer				
7	VAR_GLOBAL	IPdraglm_glob	Word(Signed)		D12	%MW0.12					
8	VAR_GLOBAL	IKphn_glob	Word(Signed)		D14	%MW0.14					
9	VAR_GLOBAL	IKlrvn_glob	Word(Signed)		D15	%MW0.15					
10	VAR_GLOBAL	IKlrvn_glob	Word(Signed)		D16	%MW0.16					
11	VAR_GLOBAL	ITlrvn_glob	Word(Signed)								
12	VAR_GLOBAL	ISetpunkt_Tl_PD_glob	Word(Signed)								
13	VAR_GLOBAL	ISetpunkt_Tl_PID_glob	Word(Signed)		D17	%MW0.17					
14	VAR_GLOBAL	ITest_Tl_glob	Word(Signed)								
15	VAR_GLOBAL	IP_glob	Word(Signed)		D18	%MW0.18					
16	VAR_GLOBAL	il_glob	Word(Signed)		D19	%MW0.19					
17	VAR_GLOBAL	IRefersanse_glob	Word(Signed)		D20	%MW0.20	settverd				
18	VAR_GLOBAL	ID_glob	Word(Signed)		D21	%MW0.21					
19	VAR_GLOBAL	IKphn_glob	Word(Signed)		D22	%MW0.22					
20	VAR_GLOBAL	IKAuto_glob	Word(Signed)		D23	%MW0.23					
21	VAR_GLOBAL	IKAuto_glob	Word(Signed)		D24	%MW0.24					
22	VAR_GLOBAL	ITAuto_glob	Word(Signed)		D25	%MW0.25					
23	VAR_GLOBAL	IAutoCtrl	Word(Signed)		D26	%MW0.26					
24	VAR_GLOBAL	INomPdrag	Word(Signed)		D27	%MW0.27					
25	VAR_GLOBAL	IKfI	Word(Signed)		D28	%MW0.28					
26	VAR_GLOBAL	IRateSek	Word(Signed)		D29	%MW0.29					
27	VAR_GLOBAL	IFlowCtrl_glob	Word(Signed)		D32	%MW0.32					
28	VAR_GLOBAL	IModus_glob	Word(Signed)		D35	%MW0.35					
29	VAR_GLOBAL	ISetpunkt_Kp_P_glob	Word(Signed)		D36	%MW0.36					
30	VAR_GLOBAL	ISetpunkt_Kp_PID_glob	Word(Signed)		D37	%MW0.37					
31	VAR_GLOBAL	ISetpunkt_Kd_PID_glob	Word(Signed)		D38	%MW0.38					
32	VAR_GLOBAL	ISetpunkt_Kp_Pl_glob	Word(Signed)		D39	%MW0.39					
33	VAR_GLOBAL	ISetpunkt_K_Pl_glob	Word(Signed)		D40	%MW0.40					
34	VAR_GLOBAL	ISetpunkt_Kp_PID_glob	Word(Signed)		D41	%MW0.41					
35	VAR_GLOBAL	ISetpunkt_Kl_PID_glob	Word(Signed)		D42	%MW0.42					
36	VAR_GLOBAL	ISetpunkt_Kd_PID_glob	Word(Signed)		D43	%MW0.43					
37	VAR_GLOBAL	ITest_Kp_glob	Word(Signed)		D44	%MW0.44					
38	VAR_GLOBAL	ITest_Kl_glob	Word(Signed)		D45	%MW0.45					
39	VAR_GLOBAL	ITest_Kd_glob	Word(Signed)		D46	%MW0.46					
40	VAR_GLOBAL	bVentil1_glob	Bit		M20	%MX0.20					
41	VAR_GLOBAL	bVentil2_glob	Bit		M21	%MX0.21					
42	VAR_GLOBAL	bVentil3_glob	Bit		M22	%MX0.22					
43	VAR_GLOBAL	bModus_glob	Bit		M23	%MX0.23	frekvensomformer eller ventil				
44	VAR_GLOBAL	bManualt_glob	Bit		M24	%MX0.24					
45	VAR_GLOBAL	bResetAlarm_glob	Bit		M25	%MX0.25					
46	VAR_GLOBAL	bAlarmCtrlLow_glob	Bit		M26	%MX0.26					
47	VAR_GLOBAL	bAlarmLow_glob	Bit		M27	%MX0.27					
48	VAR_GLOBAL	bAlarmHigh_glob	Bit		M28	%MX0.28					
49	VAR_GLOBAL	bAlarmCtrlHigh_glob	Bit		M29	%MX0.29					
50	VAR_GLOBAL	bEmergency_glob	Bit		M30	%MX0.30					
51	VAR_GLOBAL	bForoverr_glob	Bit		M31	%MX0.31					
52	VAR_GLOBAL	bStart_glob	Bit		M32	%MX0.32					
53	VAR_GLOBAL	bStopp_glob	Bit		M33	%MX0.33					
54	VAR_GLOBAL	bKlar_glob	Bit		M34	%MX0.34					
55	VAR_GLOBAL	bAutoTuneFerdig_glob	Bit		M35	%MX0.35					
56	VAR_GLOBAL	bManualtStatus_glob	Bit								
57	VAR_GLOBAL	bAlarmFerdigStatus_glob	Bit								
58	VAR_GLOBAL	bRunningStatus_glob	Bit								
59	VAR_GLOBAL	ISetpunkt_NomPdrag_glob	Word(Signed)		D47	%MW0.47					
60	VAR_GLOBAL	bVentil1Status_glob	Bit		M40	%MX0.40					
61	VAR_GLOBAL	bVentil2Status_glob	Bit		M41	%MX0.41					
62	VAR_GLOBAL	bVentil3Status_glob	Bit		M42	%MX0.42					
63	VAR_GLOBAL	IRefersanseAkt_glob	Word(Signed)		D48	%MW0.48					
64	VAR_GLOBAL	bResetAutotuning_glob	Bit		M36	%MX0.36					
65	VAR_GLOBAL	bModusEndret_glob	Bit								
66	VAR_GLOBAL	bNomPdragSatt_glob	Bit								
67	VAR_GLOBAL	IModus_savned_glob	Word(Signed)								

Label
Data Name : Master_Main
Local Label Setting

5/3/2026

	Class	Label Name	Data Type	Constant	Device	Address	Comment
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Label
Data Name : POU_01
Local Label Setting

5/3/2026

	Class	Label Name	Data Type	Constant	Device	Address	Comment
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Label

5/3/2026

Data Name : INITIERING

Function/FB Label Setting

	Class	Label Name	Data Type	Constant	Comment
1	VAR_INPUT	START	Bit		
2	VAR	STARTPREV	Bit		
3	VAR	bModus_prev	Bit		
4	VAR_INPUT	bInit	Bit		

Label

5/3/2026

Data Name : TIL_FRA_SLAVE

Function/FB Label Setting

	Class	Label Name	Data Type	Constant	Comment
1	VAR_INPUT	iReferanse	Word[Signed]		
2	VAR_OUTPUT	iTank	Word[Signed]		
3	VAR_OUTPUT	iPådragUt	Word[Signed]		
4	VAR_INPUT	bVentil1	Bit		
5	VAR_INPUT	bVentil2	Bit		
6	VAR_INPUT	bVentil3	Bit		
7	VAR_INPUT	iKp	Word[Signed]		
8	VAR_INPUT	iKi	Word[Signed]		
9	VAR_INPUT	iKd	Word[Signed]		
10	VAR_INPUT	iTf	Word[Signed]		
11	VAR_INPUT	iPådragInn	Word[Signed]		
12	VAR_INPUT	bModus	Bit		
13	VAR_INPUT	bManuell	Bit		
14	VAR_OUTPUT	FO_flow	Word[Signed]		
15	VAR	iKp_old	Word[Signed]		
16	VAR	iKi_old	Word[Signed]		
17	VAR	iKd_old	Word[Signed]		
18	VAR	iPådrag_old	Word[Signed]		
19	VAR	iTf_old	Word[Signed]		
20	VAR_OUTPUT	iP	Word[Signed]		
21	VAR_OUTPUT	iI	Word[Signed]		
22	VAR_OUTPUT	iD	Word[Signed]		
23	VAR_OUTPUT	bAlarmCritLow	Bit		
24	VAR_OUTPUT	bAlarmLow	Bit		
25	VAR_OUTPUT	bAlarmHigh	Bit		
26	VAR_OUTPUT	bAlarmCritHigh	Bit		
27	VAR_OUTPUT	bEmergency	Bit		
28	VAR_INPUT	bResetAlarm	Bit		
29	VAR_OUTPUT	iKpAuto	Word[Signed]		
30	VAR_OUTPUT	iKiAuto	Word[Signed]		
31	VAR_OUTPUT	iKdAuto	Word[Signed]		
32	VAR_OUTPUT	iTfAuto	Word[Signed]		
33	VAR	TankNy	Word[Signed]		
34	VAR	iAlarmWord	Word[Signed]		
35	VAR_INPUT	bForover	Bit		
36	VAR_INPUT	bStart	Bit		
37	VAR_INPUT	bStopp	Bit		
38	VAR_OUTPUT	iFlowLmin	Word[Signed]		
39	VAR_OUTPUT	bKlar	Bit		
40	VAR	bManuellStatus_prev	Bit		
41	VAR	bResetPuls	Bit		
42	VAR	bResetAlarm_old	Bit		
43	VAR	bEmergency_prev	Bit		
44	VAR	bStart_prev	Bit		

Label

5/3/2026

Data Name : toggle_hmi
Function/FB Label Setting

	Class	Label Name	Data Type	Constant	Comment
1	VAR_INPUT	Input	Bit		
2	VAR_OUTPUT	FlipFlop	Bit		
3	VAR	OldInput	Bit		
4	VAR	oneshot	Bit		

Label

5/3/2026

Data Name : REGULATOR_VALG

Function/FB Label Setting

	Class	Label Name	Data Type	Constant	Comment
1	VAR_INPUT	iModus	Word[Signed]		
2	VAR_INPUT	iSettpunkt_Kp_P	Word[Signed]		
3	VAR_INPUT	iSettpunkt_Kp_PD	Word[Signed]		
4	VAR_INPUT	iSettpunkt_Kd_PD	Word[Signed]		
5	VAR_INPUT	iSettpunkt_Kp_PI	Word[Signed]		
6	VAR_INPUT	iSettpunkt_Ki_PI	Word[Signed]		
7	VAR_INPUT	iSettpunkt_Kp_PID	Word[Signed]		
8	VAR_INPUT	iSettpunkt_Ki_PID	Word[Signed]		
9	VAR_INPUT	iSettpunkt_Kd_PID	Word[Signed]		
10	VAR_INPUT	iSettpunkt_Tf_PD	Word[Signed]		
11	VAR_INPUT	iSettpunkt_Tf_PID	Word[Signed]		
12	VAR_INPUT	iTest_Kp	Word[Signed]		
13	VAR_INPUT	iTest_Ki	Word[Signed]		
14	VAR_INPUT	iTest_Kd	Word[Signed]		
15	VAR_INPUT	iTest_Tf	Word[Signed]		
16					
17	VAR_OUTPUT	iKp	Word[Signed]		
18	VAR_OUTPUT	iKi	Word[Signed]		
19	VAR_OUTPUT	iKd	Word[Signed]		
20	VAR_OUTPUT	iTf	Word[Signed]		
21	VAR	bManuellStatus_prev	Bit		
22	VAR	bEmergency_prev	Bit		
23					
24	VAR	bModusEndret	Bit		
25	VAR	iModus_prev	Word[Signed]		
26	VAR	bOrganEndret	Bit		
27	VAR	bModus_prev_rv	Bit		
28	VAR	bAutoTuneFerdig_prev	Bit		